

HAYDEN FU

haydenmfu@gmail.com | (669) 236-1795 | [linkedin.com/in/hayden-fu](https://www.linkedin.com/in/hayden-fu) | haydenmfu.github.io | github.com/haydenmfu

PROFESSIONAL SUMMARY

Computer Science and Statistics student at Cornell University with experience building data pipelines and applying statistical and machine learning methods to heterogeneous, real-world datasets. Skilled in Python, probabilistic modeling, and forecasting.

EDUCATION

Cornell University, College of Arts & Sciences

Expected May 2028

B.S., Computer Science and Statistics

Relevant Coursework: Probability Models & Inference, Applied Linear Statistics, Categorical Data Analysis, Machine Learning, Algorithms, Computer Architecture, Functional Programming, Discrete Math, Linear Algebra, Multivariable Calculus

EXPERIENCE

Algoverse LLM Research Group | Applied AI Engineer

May 2026 – Present

- Built a time-aligned LLM forecasting pipeline (Python, Azure AI Foundry) combining prediction-market prices, order-book data, and news evidence to measure model calibration and drift.
- Evaluated 3 open-weight LLMs across 100 sequential forecasting tasks using linear probes and counterfactual interventions; resulting paper accepted to the IAB workshop at NeurIPS 2026.

Cornell University, Civil Engineering | Data Specialist

Jun 2026 – Present

- Built a particle-filter pipeline in Python (xarray, pandas, NetCDF, scikit-learn) for probabilistic inference over large-scale, multi-source time-series data.
- Designed preprocessing and calibration workflows to align noisy satellite observations, model outputs, and geospatial boundaries across multi-day inference windows.
- Leveraged Claude Code to debug and refactor the pipeline's diagnostics, improving readability and reproducibility.

Cornell Cybersecurity Club | Competition Team Member

Sep 2025 – Present

- Competed in international Capture-the-Flag competitions; placed 1st at BSides NYC CTF (also AmateursCTF, OSU CTF, Buckeye CTF).

Cornell Geo Data | ML Research Engineer

Nov 2025 – Present

- Built a GAN-based corrective ML model (Python, PyTorch) to improve short-term forecast accuracy using sensor and numerical prediction data.
- Cleaned and validated raw sensor time-series data to support model evaluation.

Harvard Institute of Quantitative Social Sciences | Research Intern

Jun 2024 – Aug 2024

- Applied difference-in-differences regression to housing and policy data to evaluate causal effects; published in the Journal of Student Research.

University of Illinois Urbana-Champaign, College of Education | Applied AI Developer

Sep 2023 – May 2024

- Built a data-driven map app (Python, ChatGPT-powered classification pipeline) giving K-12 educators access to 1,000+ resources.

Aspiring Scholars Directed Research Program | Independent LLM Researcher

Jan 2023 – Jun 2024

- Designed and evaluated 5 bias-mitigation prompting strategies on ChatGPT across 300+ prompts; published in JEI.

SKILLS

Languages: Python, SQL, R, Java, JavaScript, C/C++, OCaml, LaTeX

Data & ML: pandas, NumPy, scikit-learn, PyTorch, TensorFlow, matplotlib, xarray, NetCDF

Methods: Regression analysis, hypothesis testing, probabilistic inference, statistical modeling, data pipeline design

Tools: Git, Claude Code, Azure AI Foundry, Jupyter, Linux/Unix